DTC P0301:00, P0302:00, P0303:00, P0304:00 [SKYACTIV-D 1.5]

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DTC P0301:00 **DTC P0302:00** **DTC P0303:00** **DTC P0304:00**	**P0301:00: Cylinder No.1 misfire** **P0302:00: Cylinder No.2 misfire** **P0303:00: Cylinder No.3 misfire** **P0304:00: Cylinder No.4 misfire**
DETECTION CONDITION	• The misfire rate of specific cylinders relative to the crankshaft speed exceeds the specified value continuously for a specified period of time when all of the following conditions are met. — There is a record of driving the vehicle with all of the following conditions met. • Deceleration with accelerator pedal released from engine speed of **1,500 rpm** (from engine speed at deceleration start to **1,150 rpm**) • Fuel injection amount control: during fuel cut • Speed ratio fluctuation or time fluctuation for each cylinder is specified value or less — Fuel injection pattern or A/C relay operation condition is not switched directly before. — Post injection is not being performed during deceleration and no load — Gear is not being shifted — Change in accelerator opening angle is specified value or less — Speed ratio fluctuation for each cylinder is specified value or less — Fuel injection amount is specified value or more — Engine coolant temperature: **30 °C {86 °F} or more** **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits the automatic diesel particulate filter regeneration control. • Inhibits the compulsory diesel particulate filter regeneration control. • Inhibits the EGR control.
POSSIBLE CAUSE	• Out of fuel • Inferior fuel added • Leakage from fuel system • Fuel injector malfunction • Poor engine compression • Air suction or clogging into intake air system (between airflow sensor and intake manifold) • Crankshaft position sensor malfunction • Crankshaft plate damage • PCM malfunction

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY DTC CAUSING FREEZE FRAME DATA** • Is DTC P0301:00, P0302:00, P0303:00, or P0304:00 causing the freeze frame data?	Yes	Go to the next step.
		No	Inspect the DTC causing the freeze frame data. (See DTC TABLE [SKYACTIV-D 1.5].)
2	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the freeze frame data/snap shot data.	—	Go to the next step.
3	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting first, then go to the next step. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**INSPECT FOR FUEL LEAKAGE FROM FUEL LINE OR CLOGGING** • Inspect the following fuel line for fuel leakage or clogging. — Between supply pump and common rail — Between common rail and fuel injector • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 10.
		No	Go to the next step.
6	**VERIFY RELATED SENSOR INPUT SIGNAL** • Start the engine and warm it up completely. • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — ECT — IAT — MAF — RPM — APP — VSS — FRP • Are the monitoring values normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location, then go to Step 10.
7	**VERIFY INPUT SIGNAL FROM RELATED SENSOR WHILE DRIVING UNDER LOAD** • Start the engine and warm it up completely. • Drive the vehicle under the condition of the freeze frame data (mode 2) recorded in Step 2 and display the following PIDs using the M-MDS under this condition. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — ECT — IAT — MAF — RPM — APP — VSS — FRP • Are the monitoring values normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location, then go to Step 10.
8	**INSPECT FUEL INJECTORS NO.1 TO NO.4** • Inspect fuel injectors No.1 to No.4. (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) • Are the fuel injectors normal?	Yes	Go to the next step.
		No	Replace the malfunctioning fuel injector, then go to Step 10. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9	**INSPECT ENGINE COMPRESSION** • Inspect the engine compression. (See COMPRESSION INSPECTION [SKYACTIV-D 1.5].) • Is the engine compression normal?	Yes	Replace the fuel injector in all of the cylinders, then go to to the next step. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
		No	Repair or replace the malfunctioning location, then go to the next step.

STEP	INSPECTION		ACTION
10	**PERFORM FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE** **Note** • Carbon on the exhaust valve is removed by performing the fuel injector carbon deposit burn-off procedure. • Perform the "FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE". (See FUEL INJECTOR CARBON DEPOSIT BURN-OFF PROCEDURE [SKYACTIV-D 1.5].) • Is the fuel injector carbon deposit burn-off completed?	Yes	Go to Step 12.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the "AFTER REPAIR PROCEDURE". (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present? **Note** • If the fuel injector carbon deposit burn-off procedure was not successful, DTC P1284:00 is stored.	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting, then go to the next step. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
12	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle under the FREEZE FRAME DATA/snapshot data condition. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Has DTC P0301:00, P0302:00, P0303:00, or P0304:00 been stored?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM, then go to the next step. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
		No	Go to the next step.
13	**VERIFY AFTER REPAIR PROCEDURE** • Perform the "AFTER REPAIR PROCEDURE". (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.